

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine (EALSE)
Maximum Marks	2 Marks

Define Your Technology Stack

The Emotion-Aware Learning Support Engine (EALSE) is developed using a modern, scalable technology stack consisting of frontend technologies, backend frameworks, deep learning and NLP libraries, a generative AI service, data logging mechanisms, and deployment platforms. These technologies were selected for performance, maintainability, flexibility, and ease of development.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, Plotly	Provides a responsive, interactive web UI for text input, role/subject selection, result cards, and the analytics dashboard.
2	Backend / Application Logic	Python 3.10+	Handles preprocessing, model orchestration, ensemble logic, and Gemini API integration efficiently.
3	Deep Learning (Sequence Model)	TensorFlow / Keras (BiLSTM)	Provides a lightweight, fast bidirectional LSTM classifier used as a tiebreaker in the emotion ensemble.
4	Deep Learning (Transformer Model)	PyTorch + HuggingFace Transformers (DistilBERT)	Fine-tuned DistilBERT delivers the primary, high-accuracy 7-class emotion classification.
5	NLP Preprocessing	NLTK	Performs tokenisation, stopwords removal, and lemmatisation for consistent, clean text before inference.

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6	Generative AI	Google Gemini API (gemini-2.5-flash)	Generates empathetic, role- and subject-aware guidance conditioned on the detected emotion.
7	Data / Interaction Logging	CSV (Python stdlib), GoEmotions dataset	Logs every interaction locally for analytics and supplies labelled training data for the classifiers.
8	Cloud / Hosting / Deployment	Streamlit Community Cloud	Provides free, one-click deployment, easy updates, and public cloud accessibility.
9	Version Control & CI/CD	Git & GitHub	Supports version control, collaboration, and project tracking across the development team.
10	Third-Party APIs / Other Tools	python-dotenv, gdown, h5py	Manages secure API key configuration, automated model downloads, and model file serialisation.